Topic	**Groundwater Modelling**
Module	**Groundwater**
Assignment	**Planning of two new groundwater wells in the Heathcote Valley**

Due 12:00 pm on 20 March 2026 (online submission on LEARN: report in .pdf format with the filename convention Lastname_Firstname_MODr_ENCN442.pdf).
The assignment is worth 10% of the final grade.

Project description

This assignment is a planning exercise for two new groundwater wells in the Heathcote Valley area. In previous tutorials, you have developed a Heathcote Valley groundwater model. This includes a description of the topography, conductivity structure, boundary conditions, and a new water supply well. This well, BY20/0083, has been pumping at 40 L/s for the last year, and its drawdown has stabilised at about 40 m.

Two further water supply wells have been proposed for the Heathcote Valley. The intention is to drill both into a high conductivity gravel channel that is embedded in the main aquifer. A pump test analysis in the well BY20/0083 suggests that the transmissivity of the main aquifer is about 1900 m^2/d. However, no pump tests have been performed in the gravel channel, which likely has higher conductivity.

You have been provided with water levels from three observation wells screened in the gravel channel. These wells, CH5, 8 and 22, have a long-term average head of 2.5 m, 2.35 m and 2.2 m, respectively. These water levels have all declined in the year that BY20/0083 has been operating at its 40 L/s design pumpage.

Tasks

Develop further the Heathcote Valley groundwater model introduced in the tutorials. Determine a range of credible hydraulic conductivities for the gravel channel through calibration to groundwater levels.

Investigate possible sites for two new water supply wells situated in the gravel channel. When recommending their location and pumping rates, consider:

- The potential negative effects of drawdown on BY20/0083, and each other.
- Creation of hydraulic gradients that could draw in saltwater from Settlers Reserve.
- Relative ease of construction and connection into the Christchurch City Council water supply network.

Please use the starter model linked on LEARN: assignment_starter.gpt.

Report on the following:

1. Calibration: Summarise your calibration of the gravel channel hydraulic conductivity, including the preferred range of values, supported by plots that indicate why this is an acceptable range.

2. Well locations and pumping rates: Summarise your preferred location and pumping rates of the two proposed wells, supported by plots that indicate:

a) Impacts on all wells are acceptable. You should justify what is acceptable.
b) Risks of saltwater contamination are not major.

3. Modelling uncertainty: Choose ONE of the following processes NOT considered by the model: (i) pumping of other groundwater wells in Christchurch outside the model domain, (ii) upward leakage of water from aquifers deeper than those represented in the model, (iii) extreme weather or climate events (floods, sea level rise.)

For your selected process state:

a) How neglecting it creates uncertainty in the model.
b) How your model prediction, outcome, or recommendation might be different if the process were included.
c) How the process could be included in a future model.

4. Sound modelling practices: For each of the topics below, give one specific example of an action you took in relation to the Heathcote Valley model (this could include both tutorial and assignment work). For each example, state why it was important that you took this action.

a) Model conceptualisation.
b) Model development.
c) Model testing or validation.
d) Model prediction.
e) Optimisation with a model.

e.g. “Model development. I used land elevation data to create a topographically accurate surface for the model. This is important, because topography controls the direction that water flows in the model, as well as the shape of the piezometric surface.” (Do not use this example in your submission.)

Structure of the report

Note that the report should be written in 'memo-style'. This is commonly done to summarise results for internal company use (documentation of progress, update for the line manager or CEO, etc.), i.e. this is not a full client report. The memo might later become part of the larger client report. So, while this is an interim, brief document, it must still be written to the highest standard, with high quality figures and tables.

The report must be structured in the following manner:

1. Calibration
2. Well locations and pumping rates
3. Modelling uncertainty
4. Sound modelling practices

Marking scheme (100 marks total)

- Calibration: 20 marks
- Well locations and pumping rates: 40 marks
- Modelling uncertainty: 15 marks
- Sound modelling practices: 15 marks
- Report presentation: 10 marks

General remarks

- Clearly state and justify any assumption you make.
- The report must have a title, date and the name and email address of the author.
- Use adequate labels for the figures, thereby including physical units where applicable. Axes must be labeled with the name of the variable and the unit (if any). If there is more than one set of points or curves on a Figure, a legend must be added showing clearly what each one is.
- All figures must be scaled so the points, curves, etc. can be seen properly.
- Make sure that for your figures the font size has been chosen such that all numbers, labels, symbols, etc. are clearly legible.
- All figures and tables must have a numbered caption. Refer to figures and tables in the text as e.g. 'Figure 1 shows…' or '…listed in Table 1…'
- Note that a quantitative answer is more desirable than a qualitative answer. Hint: When comparing solutions, provide e.g. a percent reduction/increase in drawdown and use various types of figures and tables to present your results.
- **Page limit: 8 pages (title page does not count towards the page limit; do not include a table of contents).** Aim at keeping the report brief and to the point. You will write a great many of such reports in your future career and it is important that you practice conveying the key messages in a concise manner. Do not show all results, but those

that are relevant to effectively communicate the results. Produce meaningful and information-rich figures and tables. Despite being concise, make sure that the reader can fully understand what has been done to arrive at a particular result. For example, consider including a system schematic of the proposed solution.

- Page Margins: Minimum 20 mm
- Font and Font Size for the main body of the report: Calibri, 12 point
- Font and Font Size for Figure and Table captions: Calibri, 11 point
- Font and Font Size for section headings of the report: Calibri, 12 point, bold
- Number the pages.
- Marks will be deducted for exceeding the page limit, figures that are not legible, incorrect page margins, incorrect font size, incorrect file names, or incorrect file format.
- Lastly: There is a penalty for late submission.